General Characteristics

181. The number of incomplete orbitals in inner transition element is
(a) 2 (b) 3
(c) 4 (d) 1

182. Most common oxidation states of Cs (cesium) are
(a) + 2, + 3 (b) + 2, + 4
(c) + 3, + 4 (d) + 3, + 5

183. The $3d$ elements show variable oxidation states because the energies of the following sets of orbitals are almost similar
(a) $ns, (n-1)d$
(b) ns, nd
(c) $(n-1)s, nd$
(d) $np, (n-1)d$

184. Which of the following $3d$ bivalent metal ions has the smallest number of unpaired d electrons
(a) $3d^6$ (b) $3d^7$
(c) $3d^8$ (d) $3d^9$

185. The $3d$ metal ions form coloured compounds because the energy corresponding to the following lies in the visible range of electromagnetic spectrum
(a) Free energy change of complex formation by $3d$ metal ions
(b) $d-d$ transitions of $3d$ electrons
(c) Heat of hydration of $3d$ metal ions
(d) Ionisation energy of $3d$ metal ions

186. The oxidation number of iron in potassium ferrocyanide is
(a) +2 (b) +3
(c) +4 (d) Zero

187. In transition elements, the orbitals partially filled by electrons are
(a) *s* - orbitals (b) *p* - orbitals
(c) *d* - orbitals (d) *f* - orbitals

188. Number of unpaired electrons in Mn^{2+} is
(a) 3 (b) 5
(c) 4 (d) 1

189. Mercury is the only metal which is liquid at 0^oC. This is due to its
(a) Very high ionisation energy and weak metallic bond
(b) Low ionisation potential
(c) High atomic weight
(d) High vapour pressure

190. Essential constituent of an amalgam is
(a) Iron
(b) An alkali metal
(c) Silver
(d) Mercury

191. Mercury is transported in metal containers made of
(a) Silver (b) Lead
(c) Iron (d) Aluminium

192. The electroplating of chromium is undertaken because

(a) Electrolysis of chromium is easier

(b) Chromium can form alloys with other metals

(c) Chromium gives protective and decorative coating to the base metal

(d) Of the high reactivity of metallic chromium

193. An element having the electronic configuration $[Ar]3d^2 4s^2$ belongs to

(a) *s* - block elements

(b) *p* - block elements

(c) *d* - block elements

(d) *f* - block elements

194. Which one of the following is not a transition metal

(a) Chromium (b) Titanium

(c) Lead (d) Tungsten

195. The atomic number of an element is 22. The highest oxidation state exhibited by it in its compounds is

(a) 1 (b) 2

(c) 3 (d) 4

196. *d*-block elements form

(a) Ionic compounds

(b) Covalent compounds

(c) Ionic and covalent compounds

(d) Only complex compounds

197. The transition metals have a less tendency to form ions due to

(a) High ionisation energy

(b) Low heat of hydration of ion

(c) High heat of sublimation

(d) All of these

198. The electronic configuration of Ag atom is

(a) $[Kr]3d^{10}4s^1$

(b) $[Xe]4f^{14}d^{10}6s^1$

(c) $[Kr]4d^{10}5s^1$

(d) $[Kr]4d^9 5s^2$

199. Most powerful oxidizing property of manganese is shown by which of the following oxidation state

(a) $Mn(+7)$ (b) $Mn(+2)$

(c) $Mn(+4)$ (d) $Mn(+5)$

200. Which one of the following ions is colourless

(a) Cu^+ (b) Co^{2+}

(c) Ni^{2+} (d) Fe^{3+}